

Coding & Solution

Date	1 July 2026
Team ID	
Project Name	Opti Crop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/Sairam2220/OptiCrop-Smart-Agricultural-Production - Optimization-Engine
Programming Language(s)	Python, HTML, CSS
Framework(s) Used	Flask, Scikit-Learn
Key Features Implemented	Crop Prediction, KNN Model, Flask Web App, Responsive UI
Pending / Incomplete Features	Cloud Deployment, User Login
Setup / Run Instruction	Install requirements, Run: python app.py, Open: 127.0.0.1:5000

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes (After GitHub Push)

Additional Notes / Comments:

Code is modular, well documented, easy to maintain,
and follows clean coding practices.